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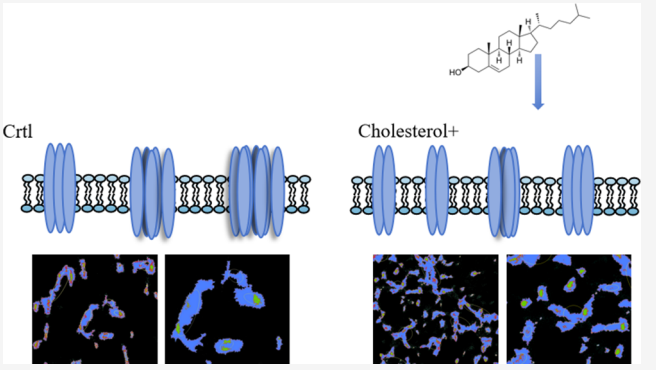
Cholesterol-Mediated Dimerization of Mitochondrial Translocator Protein TSPO: Super-Resolution Microscopy and Molecular Dynamics Insights

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ABSTRACT: The translocator protein (TSPO) is a mitochondrial membrane protein that plays a critical role in cholesterol transport and steroid biosynthesis. While its functional significance is well established, the molecular mechanisms governing TSPO dimerization, particularly the regulatory role of cholesterol, remain incompletely understood. In this study, we first employed direct stochastic optical reconstruction microscopy (dSTORM) to experimentally investigate the organization and aggregation of TSPO in living cells under different cholesterol conditions. Our imaging results revealed that cholesterol stimulation led to the formation of smaller TSPO aggregates and an increase in the point density on the cell membrane, indicating that cholesterol alters the aggregation pattern of TSPO. To further interpret these findings at the molecular level, we conducted coarse-grained molecular dynamics simulations across varying cholesterol concentrations. The simulations demonstrated that cholesterol inhibits TSPO dimerization by reducing the number of stable dimers, likely due to modifications in the local membrane environment and direct interactions with TSPO. Although the dSTORM and molecular dynamics results show differences in aggregate size and oligomerization behavior, both approaches highlight the complex regulatory role of cholesterol in TSPO organization. By integration of high-resolution imaging with computational modeling, this study provides new insights into the cholesterol-mediated modulation of TSPO dimerization, with implications for mitochondrial function and cholesterol homeostasis.



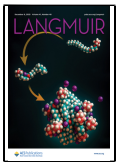
INTRODUCTION

The translocator protein (TSPO), also referred to as the peripheral-type benzodiazepine receptor (PBR), is a highly conserved cholesterol-binding protein situated in the outer mitochondrial membrane.[1−4] TSPO is widely expressed across various tissues and is present in organisms ranging from prokaryotes to eukaryotes.[1,5,6] TSPO is implicated in numerous cellular processes, including steroid biosynthesis, apoptosis, cholesterol transport, and other critical cellular functions.[7−9]

Recent structural studies using nuclear magnetic resonance (NMR) and crystallography have revealed that TSPO (Figure 1) consists of five transmembrane helices and can exist as a monomer, dimer, or oligomer, depending on the organism.[2,10−12] For example, NMR studies have demonstrated that mouse TSPO exists in a dynamic equilibrium between monomeric and dimeric states, with the dimer interface involving the TM3 region of each monomer.[13] In contrast, X-ray crystallography of bacterial TSPO,[6,14] specifically from *Rhodobacter sphaeroides*, has shown that TSPO can form both trimeric and dimeric structures (PDB IDs: 4UC2 and 4UC1, respectively). In addition, different oligomer arrangements of TSPO indicate that the functional TSPO is a dimer.[15]

TSPO's primary function is to transport cholesterol from the outer to the inner mitochondrial membrane, a crucial step in steroid and neurosteroid synthesis.[16,17] A cholesterol recognition motif (CRAC, residues L/V-(X)1-5-Y-(X)1-5-R/K) located at the cytosolic carboxyl terminus of TSPO was identified through site-directed mutagenesis and in vitro expression,[18,19] and several studies suggest that cholesterol binding to this CRAC domain may influence TSPO dimerization.[13,20,21] However, the precise mechanisms through which cholesterol regulates TSPO dimerization remain unclear, and the interface involved in dimer formation is still a subject of debate.

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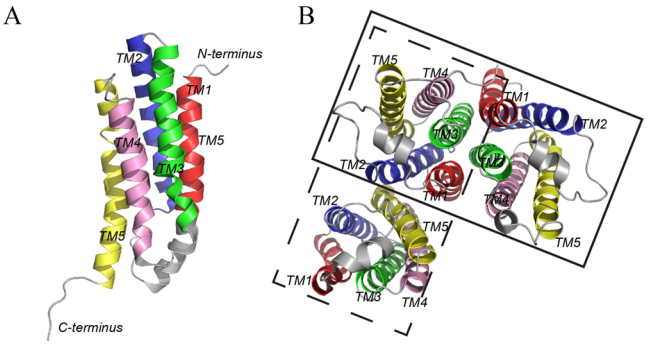


Figure 1. NMR structure and dimer of TSPO. (A) Structure of TSPO in cartoon representation (PDB ID: 2MGY). The coloring of the TM is as follows: TM1 is shown in red (VAL6-TYR24), TM2 in marine blue (TRP47-LYS69), TM3 in green (ALA78-PHE100), TM4 in pink (ALA108−127HIS), and TM5 in yellow (PRO132-TYR153). (B) Dimerization interfaces of TSPO as observed in the crystal structure (PDB ID: 4UC1).

TSPO is proposed to facilitate cholesterol translocation across the outer mitochondrial membrane via its conserved CRAC/CARC motifs and possible interactions with partner proteins such as VDAC and ANT.[22,23] These features suggest that cholesterol binding and TSPO oligomerization are functionally linked to mitochondrial cholesterol regulation. Recent studies have proposed alternative functional models for TSPO. Qiu et al. reported cholesterol-dependent enzymatic activity of human TSPO1, suggesting a potential catalytic function rather than direct cholesterol translocation.[24] While this finding represents a significant step toward understanding TSPO's multifaceted behavior, it remains debated how such enzymatic activity relates to its established role in mitochondrial cholesterol homeostasis.

To address this, we combined experimental techniques with computational simulations to explore the impact of cholesterol on TSPO dimerization. We first employed super-resolution fluorescence microscopy, specifically direct stochastic optical reconstruction microscopy (dSTORM),[25−27] to investigate the organization and aggregation of TSPO in living cells under varying cholesterol conditions. dSTORM enabled the visualization of nanoscale clustering and the distribution of TSPO on cell membranes, providing direct experimental evidence of how cholesterol modulates TSPO oligomerization. Our results showed that cholesterol induces clustering of TSPO on the membrane with noticeable changes in cluster density and size under cholesterol stimulation. These observations suggest that cholesterol may influence TSPO aggregation, but the resulting clusters may not always correspond directly to the stable cluster.

To further investigate these findings at the molecular level, we performed coarse-grained (CG) molecular dynamics (MD) simulations to examine how varying cholesterol concentrations affect TSPO dimerization. Through over a thousand simulations across different cholesterol concentrations in a POPC bilayer, we explored how cholesterol impacts TSPO dimerization patterns and its interaction with the surrounding membrane lipid environment. Our simulation results revealed increased cholesterol occupancy at transmembrane helices III and V, leading to changes in the dimeric interface of TSPO. Interestingly, we observed that higher cholesterol concentrations (10 and 30%) reduced the number of stable TSPO dimers, suggesting that cholesterol may destabilize dimer formation by altering the protein's interactions within the membrane. This observation is consistent with the clustering behavior observed in dSTORM, where cholesterol-induced changes in the distribution of TSPO led to smaller aggregates rather than to the promotion of the formation of stable clusters.

By combining these experimental and computational approaches, our study provides a comprehensive understanding of cholesterol's role in modulating TSPO oligomerization. While the dSTORM results suggest that cholesterol stimulates the clustering of TSPO on the membrane, the MD simulations indicate that cholesterol may inhibit the formation of stable TSPO dimers by affecting the protein's structural dynamics and its interaction with the membrane. These insights contribute to a more nuanced understanding of how cholesterol influences TSPO's structural dynamics and its role in mitochondrial function and cholesterol transport.

MATERIALS AND METHODS

Cell Culture. BV-2 cell was purchased from Shanghai FuHeng BioLogy. BV-2 was cultured in the DMEM medium supplemented with 10% fetal bovine serum and 1% penicillin−streptomycin in a humidified incubator at 37 °C with 5% CO_2. For confocal imaging and dSTORM imaging, the cells were cultured on a clean coverslip (22 mm × 22 mm) in a dish for at least 24 h to achieve 60−70% confluence and stimulated with 100 μM cholesterol for 12 h.

Fluorescence Labeling. After cholesterol treatment, cells were rinsed with PBS and fixed with 4% paraformaldehyde for 15 min at room temperature. Then, cells were incubated with the blocking solution (3% BSA in PBS) for 30 min at room temperature and rinsed with PBS. 0.1% Triton X-100 was added to both groups of samples and incubated at room temperature for 10 min and then washed three times with 1 mL of PBS, 3 min per wash. Staining was performed by incubating samples with the TSPO antibodies (2 μg/mL in 3% BSA; Abconal, A4881) overnight at 4 °C and washed with PBS three times. At last, samples were incubated with Alexa Fluor 647 goat antirabbit antibodies (2 μg/mL in 3% BSA; Thermo Fisher Scientific, A-21245) for 30 min at room temperature. After three washes, the cells were stained with DAPI (0.5 μg/mL; yeasen) at room temperature for 5 min for confocal imaging. 50 μL of imaging buffer containing tris (50 mM, pH 8.0), NaCl (10 mM), glucose (10% w/v), glucose oxidase (500 μg/mL; Sigma), catalase (40 μg/mL; Sigma), and β-mercaptoethanol (1% v/v; Sigma) was dropped on a large microscope slide (24 mm × 50 mm, Fisher), and the small slide (22 mm × 22 mm, Fisher) where cells were seeded was covered on the large one and sealed with nail polish. All data were acquired from ten cells in three independent experiments.

dSTORM Imaging. dSTORM imaging was performed on a Nikon Ti-E microscope with a 100 × 1.49 NA TIRF lens (Nikon, Japan). The sample was illuminated in total internal reflection fluorescence (TIRF) mode. The 640 nm laser was used to excite fluorophores of Alexa Fluor 647, and the 405 nm laser was used to increase the number of on-state fluorophores by carefully controlling its irradiation intensity (<0.1 kW/cm2). All images were captured by a cooled EMCCD (Andor Ixon Ultra 888). 5000 images were collected for each cell with an internal time of 25 ms between frames to reconstruct a super-resolution image. TetraSpeck microspheres of 100 nm (Invitrogen) were embedded in the sample to correct $x-y$ drift, and a focus lock was used to correct z drift.

dSTORM Data Analysis. Raw dSTORM image sequences were analyzed by ThunderSTORM,[28] a free available plug-in in ImageJ. The camera parameters were set before running the analysis. Then the imaging stacks were processed by a series of feature-enhancing filters to map the accurate positions of the molecules in every frame. Next, postprocessing was run to eliminate unqualified molecules with poor localization, merge reappearing molecules, and correct lateral drift. Finally, a reconstructed dSTORM image with nanoscale resolution was obtained.

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Cluster Analysis by SR-Tesseler. To characterize the spatial distribution of TSPO on cell membranes, we employed SR-Tesseler analysis.[29] First, a file containing the coordinates, intensity, and sigma values of individual localizations was loaded into the SR-Tesseler program. The reconstructed super-resolution image was displayed, and the entire cell membrane was selected as the region of interest (ROI) for analysis. Next, bisectors between the two closest localizations were drawn, segmenting the ROI into polygons, each containing a different number of localizations. The localization density of each polygon, denoted as $\delta i1$, was calculated, and the average localization density of the entire ROI, $\delta 0$, was also determined. If $\delta i1 > \delta 0$, the localizations within that polygon were identified as an "object." Similarly, the localization density of each object was defined as $\delta i2$, and the average density of all objects within the ROI was denoted as $\delta 1$. Objects where $\delta i2 > \delta 1$ were extracted as clusters. Finally, the area, number of localizations, coordinates, and other morphological parameters of each cluster were exported for further analysis. Parameters in the objects and clusters module are as follows: object segmentation: density factor = 4, min area = 2, and min #locs = 10. Cluster definition: density factor = 2, min area = 2, and min #locs = 30. A total of 10 cells were analyzed for each experimental condition. The mean values and standard errors (SE) for each group were calculated. Statistical significance between groups was assessed using an unpaired t test to determine the p-values.

MD Simulation. A series of CG simulations was conducted to investigate the dimerization of TSPO. All simulations were prepared using the docking assay for transmembrane (DAFT) components protocol[30] and performed with the Martini CG force field (MARTINI 2.2P).[31,32] The DAFT protocol facilitates the investigation of protein−protein and protein−lipid interactions by running multiple association simulations from unbiased initial states (see below). Therefore, this method enables identification of binding hotspots and alternative binding sites. To examine TSPO dimerization, three different concentrations of cholesterol were studied: 0, 10, and 30% cholesterol in 1-palmitoyl-2-oleoyl-*sn*-glycero-3-phosphocholine (POPC) lipid bilayer. All simulations were conducted using GROMACS version 2016.[33] A comprehensive summary of all of the simulation parameters is provided in Table 1.

Table 1. Summary of Simulations Performed

system	number of TSPO receptors	number of simulations	number of compact dimers	simulation length (μs)
POPC	2	500	1	5
POPC/10% cholesterol	2	500	1	5
POPC/30% cholesterol	2	472	1	5

Coarse-Grained Simulations. The initial TSPO structure was obtained from NMR data (PDB ID: 2MGY) available in the Protein Data Bank (PDB). The DAFT protocol initially converts the atomistic structures of TSPO into a CG representation using the *MARTINIZE* tool.[31] The secondary structure of the protein is maintained by applying an elastic network model (*RubberBands*) within the CG framework. The resulting CG TSPO molecules were arranged in parallel pairs and randomly rotated around the membrane normal (Z-axis) with a center-of-mass distance of 3.5 nm. For each cholesterol concentration (0, 10, and 30%), a surrounding pure POPC (approximately 350 POPC) or a mixed POPC/cholesterol bilayer (approximately 236/101 and 281/30 POPC and cholesterol molecules in 10 and 30% systems, respectively) and CG water were added using the *INSANE* tool.[34] An additional 8 CL^- ions were included to neutralize the system.

All systems were processed using the software *MARTINATE*, which first performed a 500-step steepest-descent energy minimization, followed by a 10 ps position restrained NVT simulation with a 2 fs time step. Production MD simulations, each 5 μs in duration, were performed in the NpT ensemble. The temperature was maintained at 310 K using the Berendsen thermostat, with a coupling time constant of 1 ps.[35] A semi-isotropic pressure coupling was applied using the Berendsen barostat at 1 bar, with a coupling constant of 3 ps and a compressibility of 3.0×10^{-4} bar^{-1}.[35] The Lennard-Jones interactions were shifted to zero between 0.9 and 1.2 nm, and the electrostatic interactions were switched to zero between 0 and 1.2 nm. The neighbor list for calculating short-range interactions was updated every 10 steps. Water was modeled by using the MARTINI water model with a relative dielectric permittivity of 15. An integration time step of 20 fs was used, and coordinates were written every 500 ps.

For each cholesterol condition, approximately 500 independent trajectories were simulated. However, in the 30% cholesterol system, 28 trajectories failed to equilibrate properly during the initial stages and were excluded from further analysis, resulting in 472 valid trajectories. This minor discrepancy does not affect the statistical robustness of the analysis, as all results were normalized and evaluated across the valid trajectories.

Atomistic Simulations. To further investigate the stability and structural characteristics of selected TSPO dimer conformations, key configurations (A, B2, C, D2, and E) were backmapped from CG to atomistic resolution using the backward approach.[36] The resulting atomistic models were simulated for 20 ns using a combination of the Amber-99SB force field[37] for proteins and the Slipids force field[38,39] for lipids. Prior to the simulations, the atomistic systems underwent energy minimization, followed by a 2 ns equilibration phase with positional restraints on the protein backbone atoms.

Simulations were conducted at a constant temperature of 310 K, maintained by a v-rescale thermostat[40] with a coupling time constant of 0.5 ps. Pressure was kept constant using a semi-isotropic coupling method with a Berendsen barostat,[35] employing a coupling constant of 10.0 ps. Long-range electrostatic interactions were calculated using the particle mesh Ewald (PME) method,[41] with a real-space cutoff of 1.4 nm, Fourier spacing of 0.12 nm, and fourth-order interpolation on the Ewald mesh. van der Waals interactions were handled with a cutoff at 1.5 nm using a switching function between 1.4 and 1.5 nm. Bonds involving hydrogen atoms were constrained using the LINCS algorithm[42] to ensure stability during the simulations.

Analysis. *Binding Orientation Analysis.* To determine the preferred binding orientations of dimeric TSPO structures, we analyzed data from thousands of MD simulations.[30,43,44]

The relative orientation of the two monomers was characterized using three distinct angular parameters: β, ϕ, and χ. Specifically, β represents the rotation angle of monomer B relative to the fixed reference frame of monomer A, ϕ describes the rotation of monomer B around its Z-axis, and χ denotes the relative rotational angle between monomer A and monomer B (see Figure 2). To assess the

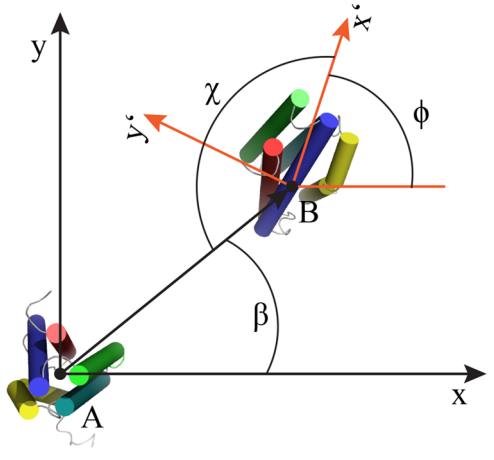


Figure 2. Relative orientation of two interacting TSPO dimers. β is the rotational angle of monomer B relative to the reference monomer A, ϕ is the rotation of monomer B around its Z-axis, and χ is the rotation angle of monomer A relative to monomer B.

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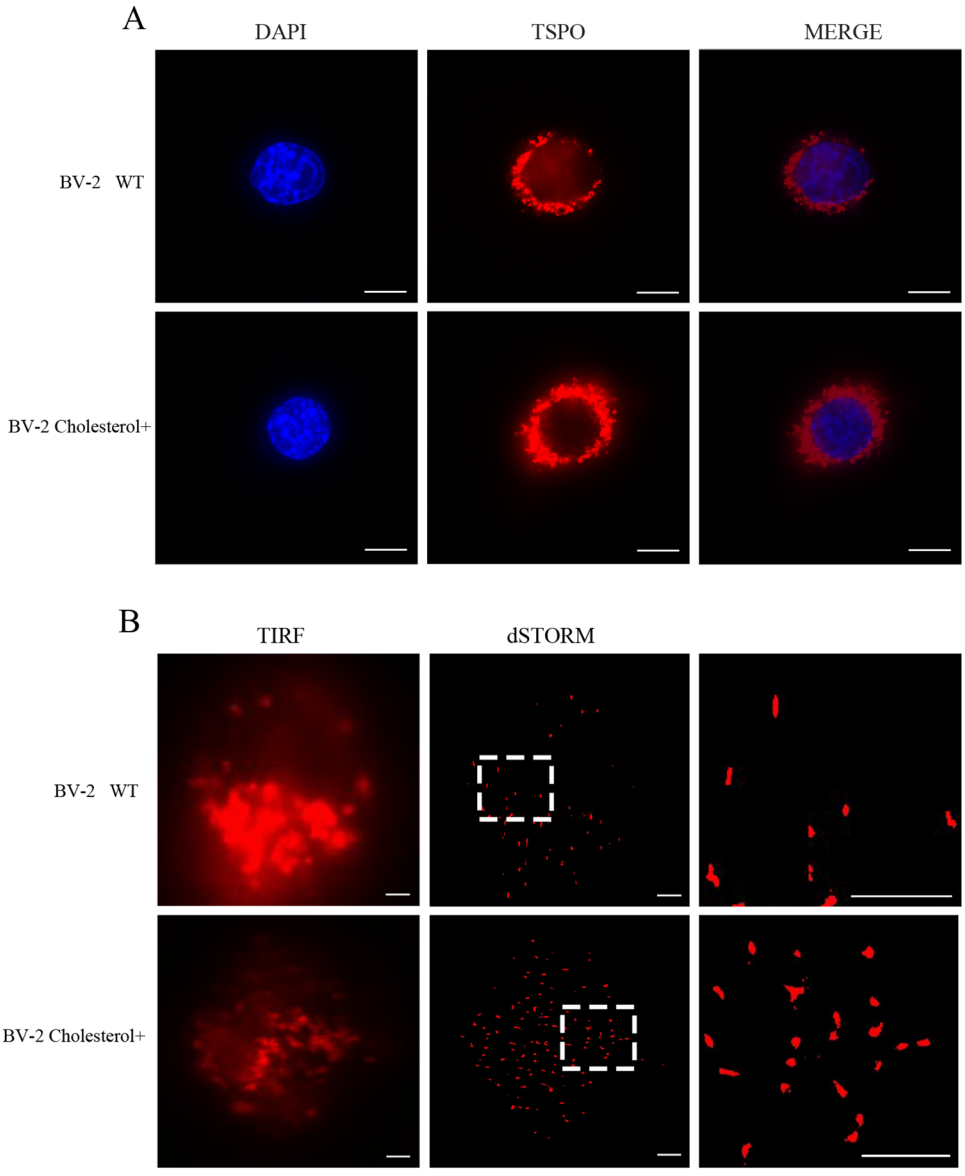


Figure 3. Morphological analysis of TSPO under cholesterol stimulation. (A) Representative confocal images of the TSPO from BV-2 cells; (B) reconstructed dSTORM images of TSPO on the membranes and the corresponding zoomed-in regions (boxed in white), with 100 μM cholesterol. Each selected image represents the most predominant one from at least 10 different cells from independent experiments under that experimental condition. Scale bars: (A) 10 μm and (B) 2 μm.

stability and distribution of these orientations, kernel density estimates were computed for the β and χ angles, utilizing data from the final 50 ns of simulation time across all spontaneously formed dimers. The probability distribution of these orientations is visually represented as a heatmap, with color gradients from blue to yellow to red indicating increasing frequency of occurrence.

Cholesterol Binding Analysis. The binding of the cholesterol molecule was defined by any amino acid residue within a 0.6 nm cutoff, as determined from the radial distribution function (RDF) calculations. In order to eliminate the random contact between cholesterol and protein, the initial 200 ns of simulations were excluded from the analysis. To identify cholesterol-binding residues, the total and maximum occupation times for any cholesterol-bound amino acid during the simulations were calculated. Total occupation time was defined as the cumulative duration of cholesterol−protein contact, while maximum occupation time referred to the longest single duration of such a contact. Both values were normalized for all simulation lengths. A similar approach was reported in.[43,45]

RESULTS AND DISCUSSION

Characterization of TSPO Cluster on Membranes of BV-2 Cells with Cholesterol Stimulation. To investigate the potential association between the cholesterol stimulation and cluster morphology of TSPO, we employ confocal imaging and dSTORM imaging with antibodies labeling BV-2 cells, a type of microglial cell that retains primary morphology, phenotype, and comprehensive functional characteristics. As shown in Figure 3A, TSPO was predominantly localized in BV-2 cells. After cholesterol stimulation, a stronger red fluorescent signal was observed, indicating that cholesterol alters the distribution of TSPO on the mitochondrial membrane. The increased fluorescence intensity likely reflects a change in the TSPO organization. Further investigation using dSTORM imaging revealed significant changes in the organization of TSPO in the mitochondrial membrane. As shown in Figure 3B, TSPO exhibited a more dispersed distribution upon

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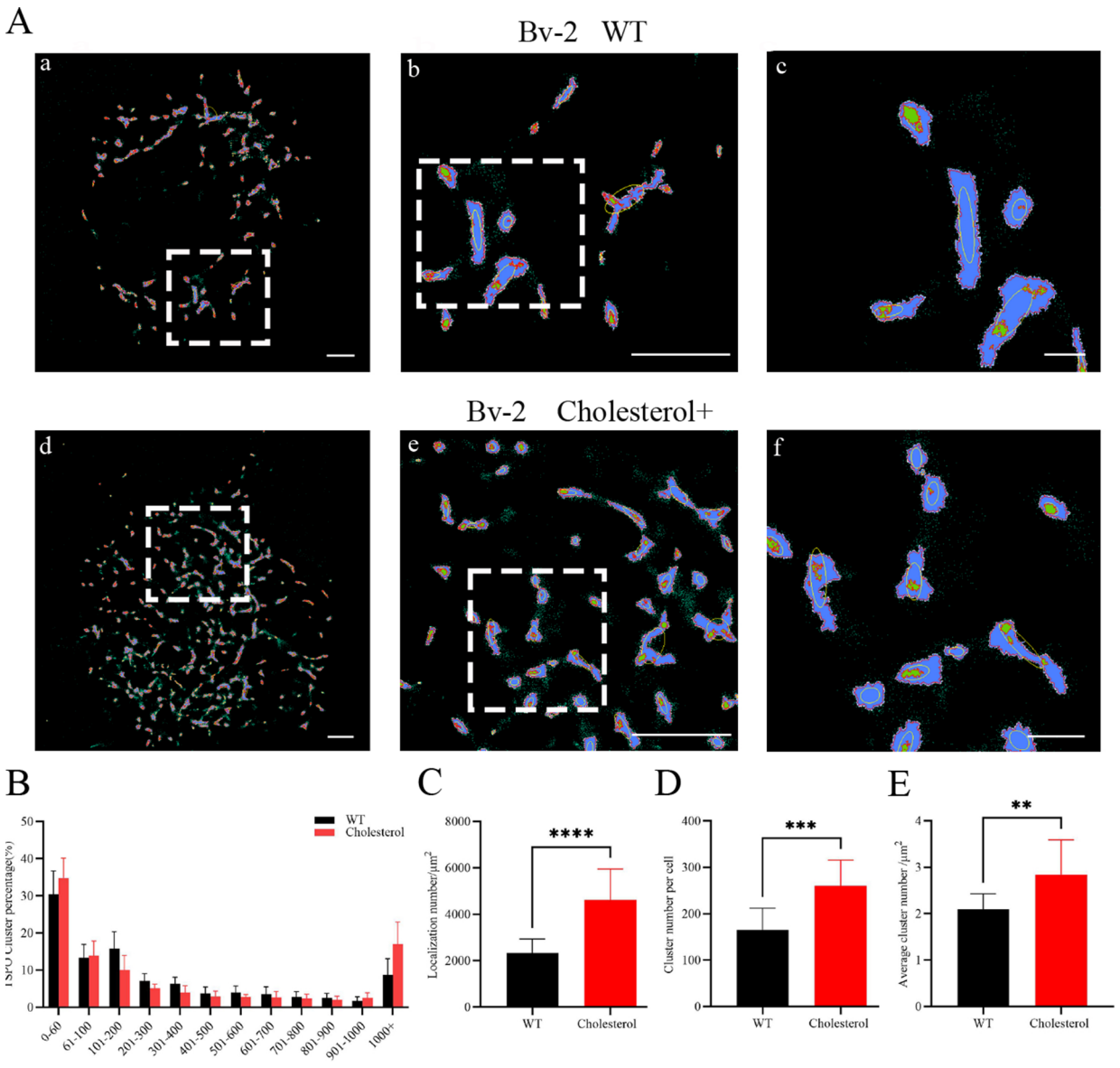


Figure 4. Comparison of cholesterol stimulation of TSPO on BV-2 cells. (A) Images of qualified clusters (green) and objects (blue) recognized by setting the appropriate thresholds with SR-Tesseler. Comparison of the distribution features of TSPO on the BV-2 cell membranes: (a–c) without cholesterol, (d–f) with 100 μM cholesterol. Scale bars: (a, b, d, e) 2 μm; (c, f) 500 nm. (B) Percentage of clusters with different cluster diameters. (C) Number of TSPO localizations per μm^2 without cholesterol, with 100 μM cholesterol. (D) Number of clusters per cell without cholesterol was 100 μM cholesterol. (E) Average cluster area per cluster (nm^2), without cholesterol and with 100 μM cholesterol. $^{**}p < 0.01$, $^{***}p < 0.001$, $^{****}p < 0.0001$. Error bars indicate means ± SD.

cholesterol treatment, with a noticeable increase in the number of TSPO molecules in closer proximity to each other. This suggests a reorganization of TSPO on the membrane, which could reflect alterations in the membrane properties due to cholesterol treatment.

We then analyzed the spatial distribution of TSPO using the SR-Tesseler analysis method, which relies on Euclidean distances to segment the membrane regions and applies multiple thresholds of localization density to identify regions of interest. The results indicated that TSPO formed clusters of varying sizes on the membranes under different conditions.[29] As shown in Figure 4A, cholesterol treatment led to an increase in the number of TSPO clusters on the membrane, with the number of clusters under cholesterol treatment being approximately 1.57 times that under the control conditions without cholesterol. Additionally, Figure 4C shows a significant increase in the point density of TSPO ($N/\mu m^2$), from 2332 before cholesterol treatment to 4633 after treatment, indicating a substantial reorganization of TSPO distribution induced by cholesterol.

We further quantified these morphological changes by analyzing several metrics, including the number of clusters and the average number of clusters per μm^2. As shown in Figure 4D,E, the total number of clusters increased from 165.5 to 260.8 per cell, while the number of clusters per μm^2 increased from 2.8 before cholesterol treatment to 2.0 after treatment. These findings suggest that cholesterol reduces the number of TSPO clusters and reorganizes the TSPO distribution on the membrane, though the average cluster size was not directly addressed in the analysis.

The observed cholesterol-dependent clustering of TSPO suggests that cholesterol modulates TSPO organization,

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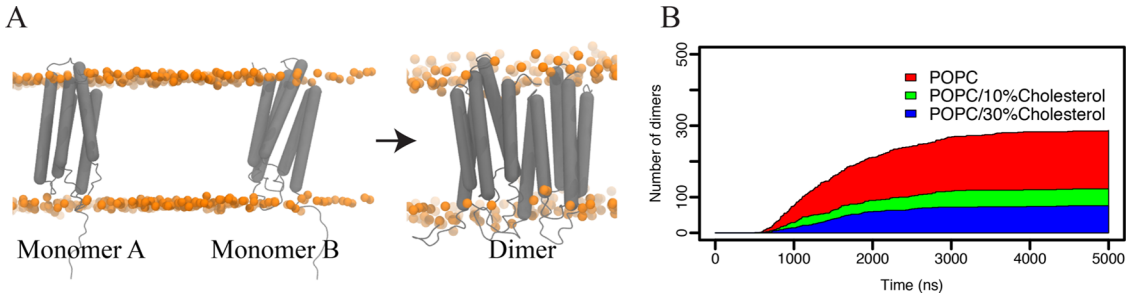


Figure 5. Spontaneous assembly of TSPO as a function of simulation time. (A) Schematic diagrams of the spontaneous assembly of two TSPO defined in all simulation systems. (B) Number of dimer configurations observed in different systems as a function of simulation time.

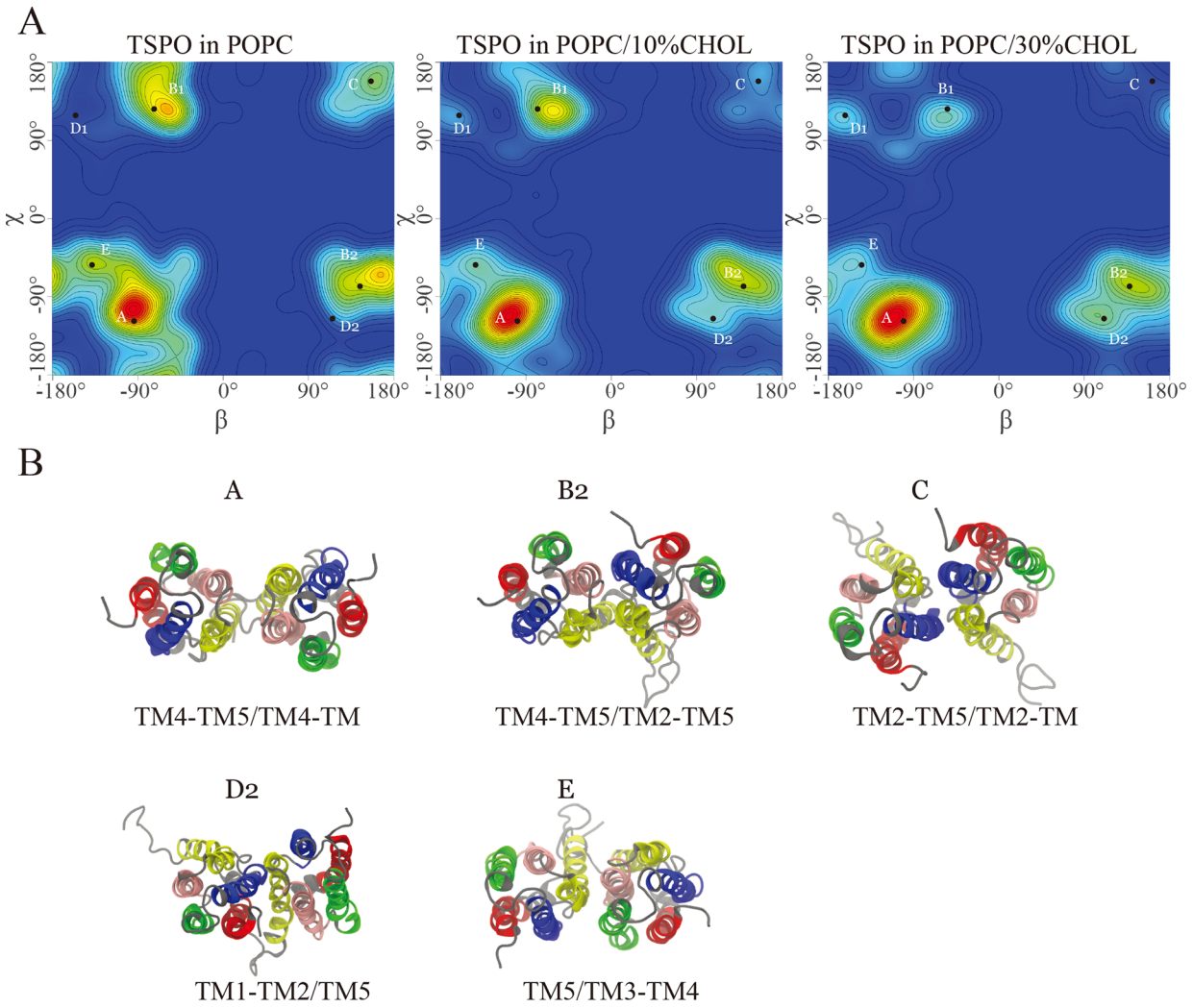


Figure 6. Cholesterol-dependent TSPO dimer configurations. (A) Sampled TSPO dimer configurations within 0, 10, and 30% cholesterol content of POPC bilayer. (B) Representative dimer configurations, labeled A through G, are depicted as helixes and are color-coded consistently with the representation in Figure 1.

potentially influencing its role in mitochondrial cholesterol transport. TSPO's ability to bind and translocate cholesterol across the mitochondrial membrane could be altered by changes in its oligomeric state. Moreover, cholesterol-induced TSPO aggregation may also influence its interaction with other mitochondrial proteins, such as VDAC and ANT, which are involved in mitochondrial membrane dynamics and energy production. These findings highlight the importance of TSPO's structural dynamics in its biological functions, which warrants further investigation.

In mammalian cells, approximately 60−80% of total cholesterol is localized in the plasma membrane, where it plays a key role in regulating membrane order and protein organization. Considering that cholesterol can modulate the conformation and aggregation state of TSPO, the plasma membrane of BV-2 microglial cells provides a feasible in vitro model to explore the 'cholesterol−TSPO−mitochondria' axis.

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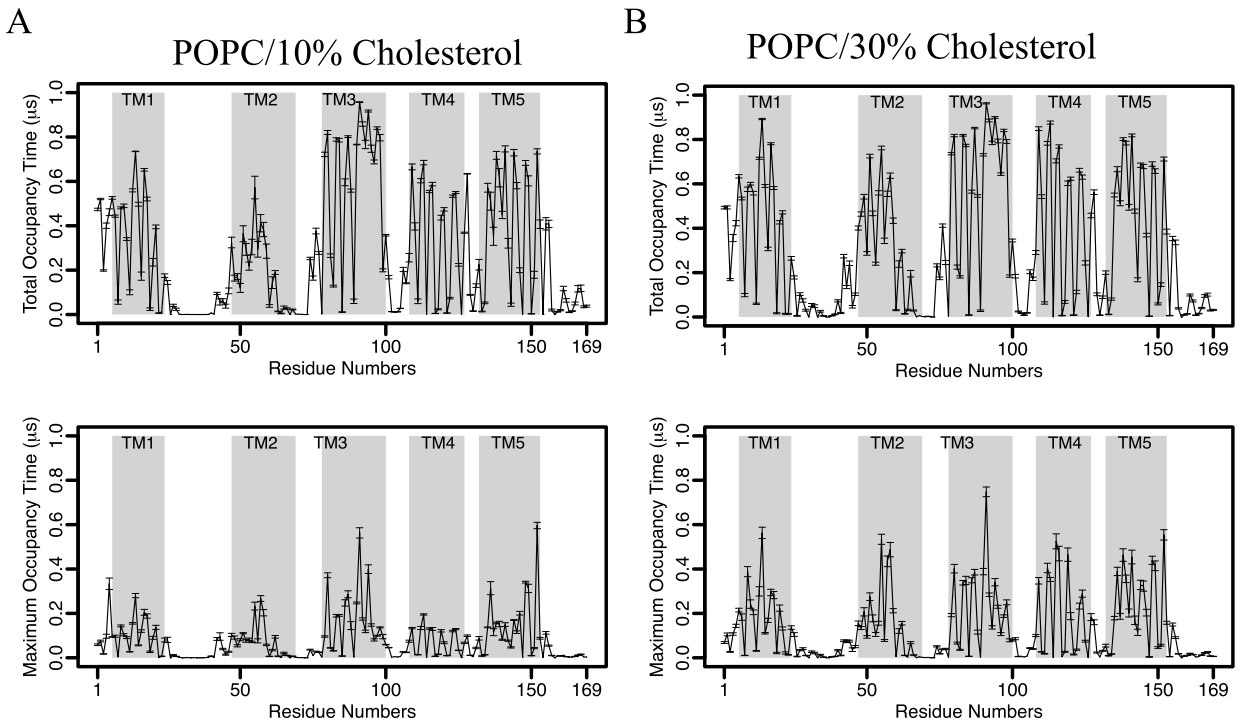


Figure 7. Cholesterol binding occupancy at each residue of TSPO in POPC bilayers with different cholesterol concentrations. (A) Total and maximum occupancy times of cholesterol at each residue of TSPO in a POPC bilayer with 10% cholesterol content, averaged over all simulations. (B) Total and maximum occupancy times of cholesterol at each residue of TSPO in a POPC bilayer with 30% cholesterol content, averaged over all simulations.

This system also serves as a practical platform for pharmacological ligand screening and mechanistic validation.[46−48]

Assembly of TSPO Dimer. The dimerization behavior of TSPO in POPC bilayers containing 0, 10, and 30% cholesterol was investigated by using CG MD simulations. To ensure statistical robustness, 500 independent simulations were conducted for each system. The number of TSPO dimers was analyzed by calculating the minimum distance between two TSPO monomers as a function of time for each simulation (data not shown). A TSPO dimer was defined based on an intermonomer distance cutoff of 0.5 nm, as referenced in previous studies.[49,50] Initially, in each simulation, the two TSPO monomers diffused freely, maintaining an average separation distance of approximately 4.2 nm. After 5 μs of simulation, the number of dimers observed was found to be dependent on the cholesterol concentration. The presence of cholesterol resulted in a reduced level of dimerization, as illustrated in Figure 5. Specifically, at cholesterol concentrations of 10 and 30%, fewer dimer species were observed compared to the pure POPC bilayers, suggesting that cholesterol alters TSPO dimerization by either modifying the membrane properties or directly interacting with TSPO. This finding is consistent with the results of Guo et al.,[6] which demonstrated that the TSPO oligomeric state is highly dependent on the surrounding lipid environment, suggesting that membrane composition can significantly influence the assembly and stability of TSPO dimers. In addition, long-lived stable dimers and transient dimers were both observed in the simulations.

These findings complement the results from our dSTORM imaging, in which cholesterol treatment led to the clustering of TSPO on the membrane. While the dSTORM results indicate an increase in the number and size of TSPO clusters under cholesterol stimulation, the MD simulations suggest that cholesterol destabilizes stable TSPO dimers. This apparent discrepancy can be interpreted as follows: the larger aggregates observed in the dSTORM images may not correspond to stable, functional dimers but rather reflect transient or loosely bound TSPO molecules that are influenced by cholesterol-induced changes in the membrane environment.

Cholesterol Modulates the Dimer Interface of TSPO. To gain a deeper understanding of how cholesterol affects the dimerization of TSPO, we analyzed the relative orientations of TSPO monomers in all dimer structures across each system. These conformations were determined by measuring the rotational angles between two TSPO monomers relative to each other (see Analysis and Figure2 for details). Seven distinct dimer conformations were identified, as shown in Figure 6A. In pure POPC bilayers, the most prevalent conformation was Conformer A, representing a compact asymmetric dimer involving the TM4/TM5 interface. The second most populated conformation, Conformer B, featured a symmetric TM2-TM5/TM4-TM5 interface. A symmetric TM2/TM5 interface was also observed in Conformer C (Figure 6B). The flexibility of TSPO monomers allows for rotation around each other, resulting in a variety of dimer interfaces, with Conformer A showing high prevalence across all systems analyzed.

When cholesterol was introduced into the POPC bilayer, significant shifts in the population of the dimer conformations were observed. In the mixed POPC:Cholesterol bilayer, the prevalence of Conformers B and C decreased, while related Conformers D1 and D2 emerged. Interestingly, at higher cholesterol concentrations (10 and 30%), instances of Conformer E were also observed. These changes suggest that

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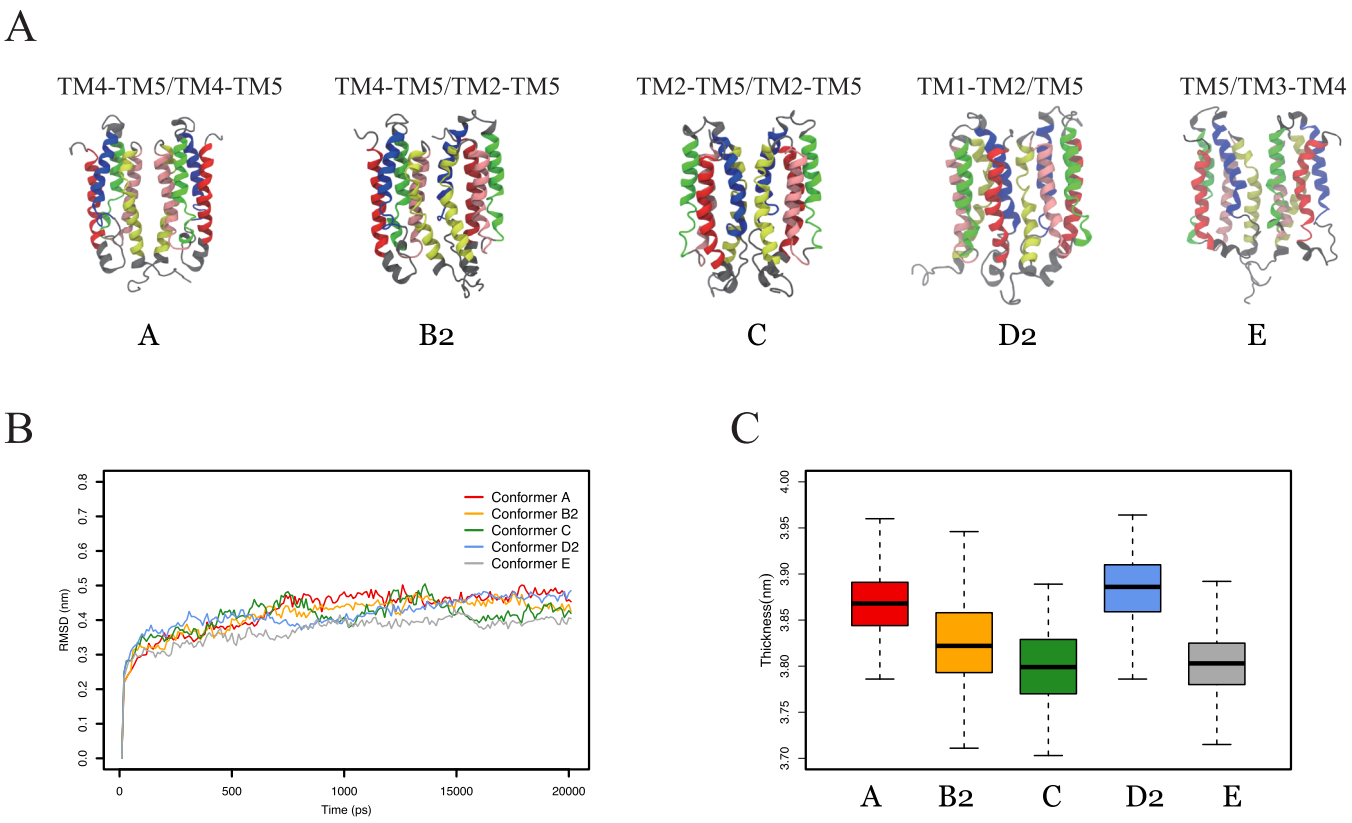


Figure 8. Analysis of TSPO dimer configurations after atomistic simulations. (A) Final snapshots of spontaneously formed TSPO dimers from conformations A, B2, C, D2, and E. Structures were obtained from the last snapshot of the 20 ns atomistic simulations of each dimer configuration, following backmapping from CG to atomistic resolution. (B) RMSD of each TSPO dimer conformation over 20 ns atomistic simulations. RMSD values indicate the stability of each conformation during the simulation period. (C) Membrane thickness analysis for each TSPO dimer conformation.

cholesterol profoundly influences the structural dynamics of TSPO dimerization, likely by altering the membrane's properties, such as fluidity and thickness, or through direct interactions with TSPO monomers.

The shift in dimer populations observed in the MD simulations aligns with the dSTORM results, where cholesterol treatment promoted clustering but did not necessarily stabilize TSPO dimers. The changes in the dimer interfaces further suggest that cholesterol alters TSPO's ability to adopt different conformations, potentially by modifying the flexibility of the monomers or the lipid environment surrounding the protein. This finding is consistent with previous work by Jaipuria et al.,[13] who reported that cholesterol can induce allosteric changes in membrane proteins, thereby influencing their dimerization and function.

Previous studies, such as those by Rao et al. and Lai et al., have investigated TSPO dimerization using CG simulations, revealing that cholesterol modulates the dimerization process.[51,52] Our findings align with these studies have shown that cholesterol affects TSPO oligomerization. However, our combined experimental and computational approach, using dSTORM imaging and DAFT simulations, offers more statistically reliable insights into TSPO clustering and aggregation, providing experimental validation and a more comprehensive understanding of the cholesterol-mediated modulation of TSPO dimerization.

To further characterize the amino acid residues involved in cholesterol binding, the total and maximum occupancy times of cholesterol molecules at each residue were calculated over all of the simulations. As illustrated in Figure 7, the cholesterol occupancy was consistently highest at transmembrane helix III (TM3) in both 10 and 30% cholesterol systems. This finding indicates a strong affinity of TM3 for cholesterol binding under different cholesterol conditions. The total occupancy data revealed that cholesterol interacts with all transmembrane regions (TM1 to TM5), with notably high occupancy (>0.8) at helices I (TM1), III (TM3) and V (TM5). The maximum occupancy times, representing the longest duration of cholesterol binding, were also primarily centered on TM3, highlighting its potential role in stabilizing cholesterol interactions.

Residues Val14, Thr55, and Leu91, corresponding to helices I, II, and III, respectively, exhibited the highest maximum occupancy times, suggesting these residues are critical for cholesterol binding and may influence TSPO's structural stability. The consistent binding patterns across different cholesterol concentrations suggest that TM3, along with key residues in TM1 and TM5, may be crucial for TSPO's interaction with cholesterol and its role in cholesterol transport. While TM1 and TM5 dominate interfacial contacts, TM3 appears as an ancillary contributor whose contact frequency increases under higher cholesterol but remains less pronounced than that of TM1/5. In the CG ensemble, recurring dimer interfaces were mainly formed through TM4−TM5 and TM2−TM5 contacts, with TM5 residues adjacent to CRAC/CARC motifs frequently involved in the interaction. This arrangement is compatible with cholesterol-sensitive

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organization that may modulate TSPO clustering and structural stability.

Stability of TSPO Dimers and Membrane Thickness. To further explore the structural characteristics and environmental interactions of TSPO dimers, selected conformations (A, B2, C, D2, and E) were converted from CG to all-atom representations and simulated for 20 ns. Figure8B presents the root mean square deviation (RMSD) for each TSPO dimer conformation over a 20 ns simulation period. All trajectories rapidly converged within the first few nanoseconds and remained stable around 0.4−0.5 nm throughout the simulations, indicating overall structural retention under atomistic conditions. Minor fluctuations were observed among conformations but were within the expected range for independent trajectories. These results suggest that the dimer interfaces obtained from CG simulations remain structurally stable after conversion to all-atom models.

Figure8C shows the membrane thickness around each dimer conformation during the simulations, calculated using the FATSLiM tool.[53] Conformations A and D2, which demonstrated higher stability and lower solvent exposure, were also associated with an increased membrane thickness in their vicinity. This suggests that these stable dimer configurations might induce or require thicker lipid bilayers, potentially due to specific lipid−protein interactions that reinforce dimer stability. On the other hand, conformations B2 and C, which showed higher structural flexibility and solvent exposure, were associated with thinner membranes, indicating that these dimer interfaces might prefer or promote a different lipid environment, possibly more disordered or less tightly packed.

CONCLUSIONS

In this study, we explored the effect of cholesterol on TSPO aggregation by using both experimental dSTORM imaging and MD simulations. Our findings reveal that cholesterol promotes TSPO clustering on the membrane, as observed by dSTORM, but simultaneously reduces the formation of stable TSPO clusters, as shown by MD simulations. The simulations indicate that cholesterol modulates the dimerization interface by affecting specific regions, particularly the TM4-TM5 and TM1-TM5 interfaces, and alters membrane properties such as fluidity and thickness. These changes in the lipid environment influence TSPO's ability to adopt different dimer conformations, enhancing dimer stability at lower cholesterol concentrations. Additionally, key residues in TM1, TM3, and TM5 were identified as critical for cholesterol binding, suggesting that cholesterol directly interacts with TSPO to modulate its functional state. Our study provides a comprehensive understanding of the molecular mechanisms underlying TSPO dimerization and its regulation by cholesterol, offering new insights into its role in cholesterol transport and mitochondrial function. These findings open up avenues for future studies targeting TSPO in cholesterol-related diseases and mitochondrial disorders.

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Notes

The authors declare no competing financial interest.

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